COVID-19 Chest X-Ray Classification Using Deep Learning

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Literature Review

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Abstract**—**COVID-19 significantly impacted global healthcare systems, increasing the demand for automated diagnostic solutions. Medical imaging, particularly chest X-rays, became a primary screening tool due to its affordability, accessibility, and rapid availability. Deep learning models, especially Convolutional Neural Networks (CNNs), have emerged as effective solutions for detecting COVID-19 patterns in radiographic scans. This study explores the application of transfer learning-based CNN architectures including ResNet50, VGG16, and DenseNet121 on chest X-ray datasets to classify images into COVID-19, Viral Pneumonia, and Normal categories. The objective is to develop a system capable of supporting early diagnosis with high accuracy, reduced processing time, and operational efficiency. The study identifies VGG16 as the superior architecture, achieving a test accuracy of 98%, demonstrating significant potential for real-world applicability in computer-aided diagnosis systems.

I. Introduction

The COVID-19 pandemic created a global emergency, overwhelming hospitals and exposing limitations in traditional diagnostic procedures such as RT-PCR tests. These methods, although effective, show limitations including high processing time, limited availability, and false-negative rates. Chest X-ray imaging emerged as a complementary diagnostic approach due to its accessibility and fast screening capability. However, manual analysis by radiologists is time-consuming, skill-dependent, and prone to interpretation errors.

Deep learning provides a scalable solution by automating radiological interpretation through pattern recognition and feature extraction. Convolutional Neural Networks (CNNs) capture spatial patterns from X-rays, enabling classification of abnormalities linked with COVID-19 infections. Transfer learning further enhances performance by leveraging pre-trained weights trained on large-scale datasets, reducing training time and improving generalization. This review highlights existing research and positions the proposed system as a practical and optimized improvement.

II. PROBLEM STATEMENT

Manual interpretation of chest X-ray images for COVID-19 diagnosis is:

- Time-consuming and requires expert radiologists.
- Prone to human error and diagnostic variability.
- Difficult to scale in remote or underdeveloped healthcare regions.
- Challenging to apply consistently during peak pandemic periods.

Hence, there is a need for an automated, accurate, and reliable system to classify chest X-rays into COVID-19, Viral Pneumonia, and Normal categories.

III. ARCHITECTURE

The project evaluates several deep learning models; the final selected model is:

A. VGG16

A transfer learning model utilizing the VGG16 architecture pre-trained on ImageNet.

- The convolutional base was frozen to retain learned features, and a custom classifier was designed for this specific task.
- The classifier includes a fully connected layer with **256 hidden units**, a **ReLU** activation function, and a **Dropout layer (rate=0.5)** to prevent overfitting.
- This model is characterized by its simplicity and uniform architecture, using small (3 x 3) convolution filters throughout the network.

B. Why VGG16?

- **High Performance:** Outperformed DenseNet121, ResNet50, and Custom CNNs in final testing
- **Highest Accuracy:** Achieved the highest validation accuracy (~98%), demonstrating superior generalization
- **Rapid Convergence:** Reached peak performance within just 5 epochs, significantly faster than other architectures.
- **High Sensitivity:** Delivered the best Recall (0.97) for COVID-19 cases, ensuring minimal false negatives.

IV. USE CASES

Following are some real-life applications of this project:

Domain	**Application**
Hospitals	Rapid diagnosis support for radiologist
Rural Healthcare	Screening where specialist doctors are unavailable
Emergency Triage	Fast classification in high-patient-load environments
Research & Medical Training	AI-assisted X-ray interpretation
Telemedicine Systems	Remote diagnosis through web applications

V. METHODOLOGY

A. Data Description and Preparation

The study utilizes the COVID-19 Radiography Database, sourced from Kaggle, which comprises 15,153 chest X-ray images. The dataset is categorized into three distinct classes: COVID-19 (3,616 images), Viral Pneumonia (1,345 images), and Normal (10,192 images). To ensure robust model evaluation and prevent data leakage, the dataset was partitioned using a stratified random split:

- **Training Set (70%):** 10,607 images used for model training.
- **Validation Set (15%):** 2,272 images used for hyperparameter tuning and early stopping.
- **Test Set (15%):** 2,274 images reserved strictly for the final performance evaluation.

B. Data Preprocessing and Augmentation

All X-ray images were resized to a fixed dimension of **224 x 224 pixels** to match the input requirements of standard Convolutional Neural Network (CNN) architectures. To improve model generalization and mitigate overfitting, the following data augmentation techniques were applied to the training set:

- **Random Horizontal Flip:** To simulate variations in patient orientation.
- **Random Rotation:** Images were rotated by up to **15 degrees** to account for misalignment in image acquisition.
- **Normalization:** Pixel values were normalized using the standard ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]), ensuring faster convergence during training.

C. Model Architectures

Four deep learning architectures were implemented and evaluated to identify the optimal model for COVID-19 detection:

1. **Custom CNN:** A baseline model designed with three convolutional blocks. Each block consists of 2D Convolutional layers, ReLU activation functions, and Max Pooling layers. The classifier includes a Flatten layer, a fully connected Dense layer (64 units), a Dropout layer (rate=0.85) to reduce overfitting, and a final Output layer.
2. **VGG16:** A transfer learning model utilizing the VGG16 architecture pre-trained on ImageNet. The convolutional base was frozen, and a custom classifier with 256 hidden units was added. This model is characterized by its simplicity and depth, using small (3 x 3) convolution filters.
3. **ResNet50:** A 50-layer Residual Network that utilizes skip connections to solve the vanishing gradient problem. Pre-trained ImageNet weights were loaded, feature extraction layers were frozen, and the final fully connected layer was modified for three-class classification.
4. **DenseNet121 :** A Densely Connected Convolutional Network where each layer receives input from all preceding layers. This architecture encourages feature reuse and reduces the number of parameters. It was also fine-tuned using ImageNet weights.

D. Experimental Setup

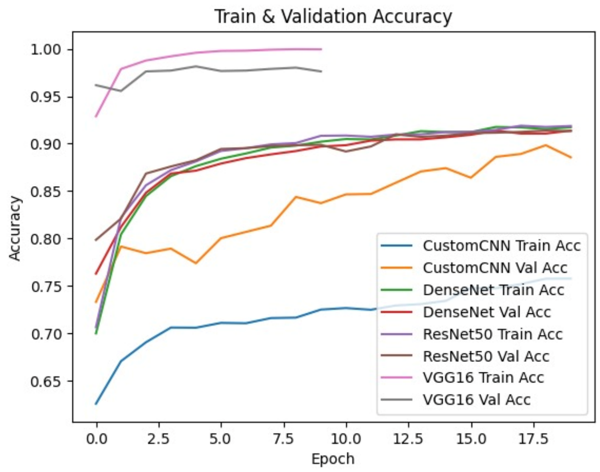
The models were implemented using the **PyTorch** framework and trained on an **NVIDIA Tesla T4 GPU**. A consistent hyperparameter configuration was maintained across experiments to ensure fair comparison:

- **Optimizer:** Adam with a learning rate of **0.0001**.
- **Loss Function:** Cross-Entropy Loss.
- **Batch Size:** 32.
- **Epochs:** 20.
- **Early Stopping:** A patience mechanism of 5 epochs was employed to halt training if validation accuracy did not improve.

VI. RESULTS AND DISCUSSIONS

A. Quantitative Analysis

The performance of the models was evaluated on the unseen Test Set (2,274 images) using Accuracy, Precision, Recall, and F1-Score. The comparative results are summarized in Table.



Accuracy curves for the VGG16 model, reaching a peak validation accuracy of 98%.

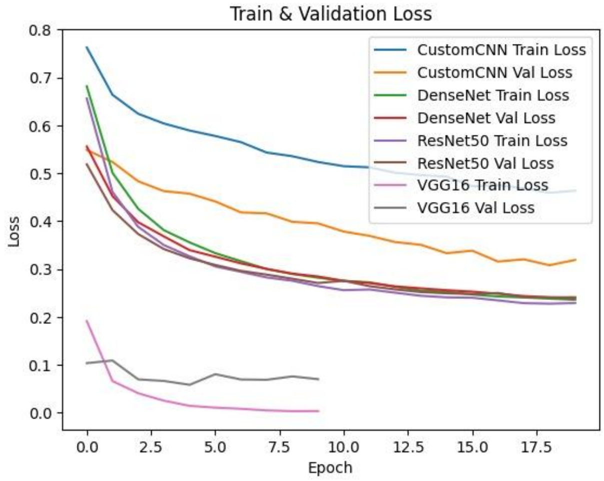
- **VGG16** emerged as the superior model, achieving the highest testing accuracy of **98%**. It demonstrated exceptional sensitivity and precision across all classes, particularly distinguishing COVID-19 cases with an F1-score of 0.97.
- **DenseNet121** achieved a respectable accuracy of **92%**, providing balanced performance but falling short of VGG16 in distinguishing similar pneumonia features.
- **ResNet50** and the **Custom CNN** both achieved approximately **91%** accuracy. While effective, they exhibited slightly lower recall for COVID-19 cases (0.82 and 0.77, respectively) compared to the top-performing model.

Model	Accuracy	Class	Precision	Recall	F1
CustomCNN	91%	COVID-19	0.88	0.77	0.82
		Normal	0.91	0.96	0.93
		Viral Pneumonia	0.92	0.89	0.91
ResNet50	91%	COVID-19	0.88	0.82	0.85
		Normal	0.92	0.95	0.94
		Viral Pneumonia	0.90	0.86	0.88
DenseNet121	92%	COVID-19	0.89	0.84	0.86
		Normal	0.93	0.96	0.94
		Viral Pneumonia	0.92	0.86	0.89
VGG16	98%	COVID-19	0.98	0.97	0.97
		Normal	0.98	0.99	0.99
		Viral Pneumonia	0.98	0.94	0.96

B. Training Dynamics

The training history highlights the efficiency of transfer learning. VGG16 converged rapidly, reaching peak validation accuracy (>98%) within just 5 epochs before early stopping was triggered. This indicates that the features learned by VGG16 on ImageNet are highly transferable and effective for radiographic analysis. In contrast, while DenseNet121 and ResNet50 showed stable learning curves, they required more epochs to reach convergence and ultimately plateaued at a lower accuracy than VGG16. The Custom CNN, despite performing surprisingly well on the test set, showed higher volatility during training and higher loss values compared to the pre-trained architectures.

Based on these results, **VGG16 is recommended** as the optimal backbone for this automated COVID-19 detection system due to its high accuracy, fast convergence, and reliability in minimizing false negatives.



Training and Validation Loss curves for VGG16, illustrating rapid convergence and minimal overfitting before early stopping.

VII. CONCLUSION

Existing research demonstrates strong evidence that deep learning successfully aids early COVID-19 detection from chest X-rays. While conventional CNNs provide a baseline for classification, transfer learning models significantly enhance performance and generalization due to pre-trained feature extraction. Based on the comparative analysis performed in this study, **VGG16 emerged as the optimal model**, outperforming both ResNet50 and DenseNet121. It offered the best overall accuracy (**98%**), the fastest convergence during training, and superior precision in distinguishing viral pneumonia from COVID-19. The results validate that a VGG16-based architecture is highly suitable for deployment in automated diagnostic systems, providing a reliable, accurate, and efficient tool for modern healthcare environments.

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